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Primary Research Area: Natural Language Processing (NLP)

Secondary Research Area: Generative AI

Summary of the Project work

1. Introduction to the Domain

Long Document Summarization (LDS) aims to condense lengthy, information-rich documents—such as research papers, patents, and government reports—into concise summaries that retain essential information. Unlike short document summarization, LDS presents additional challenges:

- **Length Constraints:** Documents often exceed thousands of tokens, going beyond the input limits of standard models.
- **Breadth of Content:** Summaries cannot cover all details, increasing the likelihood of information loss.
- **Coherence Requirements:** Summaries must maintain logical flow and semantic consistency across sections.

LDS has applications in research, legal, business, and policy-making domains, enabling faster comprehension and decision-making.

2. Introduction to Transformers in LDS

Transformers, with their attention-based architecture, have transformed the NLP landscape and are now adapted for LDS. Originally designed for shorter texts, their adaptation for long documents involves techniques such as:

- **Efficient Attention Mechanisms** (e.g., Longformer, BigBird) to handle longer inputs.
- **Content Selection Strategies** to pre-filter important sections before summarization.
- **Hybrid Models** combining extractive and abstractive summarization.

These advancements allow models like BART and PEGASUS to handle complex, domain-specific language in LDS tasks.

3. Base Paper Details

Title: An Empirical Survey on Long Document Summarization: Datasets, Models, and Metrics

Authors: Huan Yee Koh, Jiaxin Ju, Ming Liu, Shirui Pan

Journal: ACM Computing Surveys (SCIE Indexed)

Year: 2022

Official Link: https://doi.org/10.1145/3545176

Journal Homepage: https://dl.acm.org/journal/csur

Summary:

The survey provides an in-depth analysis of LDS, covering:

- **Datasets:** Characteristics and comparisons of major LDS datasets such as arXiv, PubMed, BIGPATENT, GovReport, and BillSum.
- **Models:** Evolution from graph-based extractive models to advanced Transformer-based and hybrid architectures.
- **Evaluation Metrics:** ROUGE, BERTScore, semantic coherence, and informativeness coverage.
  It also discusses dataset noise issues, model limitations, and future research opportunities, including controllable summarization and multi-metric evaluation.

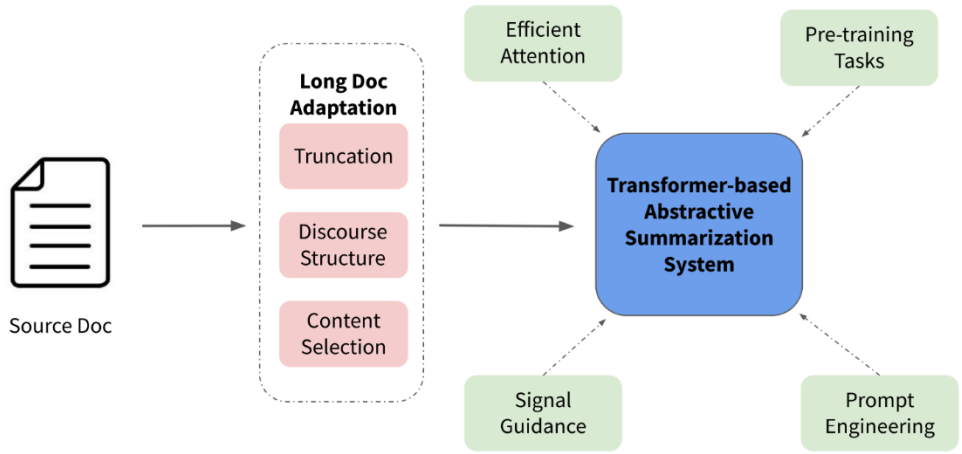

Fig. 3. Overview of transformer-based abstractive and hybrid summarization models.

4. Sample Student Project Titles

- **Efficient Transformer Architectures for Long Document Summarization – Compare Longformer and BigBird performance on arXiv and PubMed.**
- **Noise-Resilient Summarization Models – Address data quality issues in LDS datasets.**
- **Hybrid Summarization with Content Selection – Improve coherence by pre-selecting salient content.**
- **Domain-Specific LDS Systems – Specialize models for patents, legal, or financial documents.**
- **Beyond ROUGE – Design an LDS evaluation framework incorporating multiple metrics.**

5. Datasets

Recommended publicly available datasets:

- **arXiv** – Scientific research articles.
- **PubMed** – Biomedical literature.
- **BIGPATENT** – Patent documents with human-written summaries.
- **GovReport** – Government accountability reports.
- **BillSum** – U.S. legislative bills.

6. Expected Outcomes

By project end, students should:

- **Understand LDS challenges and techniques.**
- **Implement and evaluate a Transformer-based or hybrid LDS system.**
- **Compare datasets and analyze model trade-offs.**
- **Explore advanced mechanisms such as efficient attention or discourse bias.**
- Document findings, challenges, and ideas for future LDS improvements.

Students should aim for innovation and real-world applicability rather than replicating baseline results.

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